

A Self-Polishing Coating with Dual Anti-Fouling Strategies and its Preparation Method

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Anti-fouling coatings are an effective strategy for combating biofouling in the marine industry. However, traditional anti-fouling coatings have the disadvantages of high toxicity and unstable polymer degradation rates, which impact on the environment and economy. Herein, a self-polishing, anti-fouling coating without heavy-metal biocides and with dual anti-fouling mechanisms is prepared. By grafting N-(4-hydroxy-3-methoxybenzyl), which is a repellent effect on fouling organisms, onto an acrylic resin and introducing polydimethylsiloxane chains to reduce surface energy, the adhesion of *Staphylococcus aureus*, *Vibrio alginolyticus* and *Nitzschia sp.* on the coating surface is reduced by $\approx 90.2\%$, $>99.9\%$ and $\approx 89.8\%$, respectively. To ensure that the coating can form a self-renewing surface under seawater scouring, degradable polycaprolactone chain is used to bridge the acrylic resin; thus, the mechanical properties of the coating are improved. This study provides a solution to the problem of biofouling in the marine industry, biomedicine, and other fields.

coatings.^[2] These copolymers do not have anti-fouling properties; however, they are continuously polished off during ship voyage, releasing the heavy-metal biocides and effectively inhibiting the adhesion of marine fouling organisms.^[3] However, heavy-metal biocides used in self-polishing anti-fouling coatings lack specificity and exhibit high toxicity to fouling organisms and non-target marine organisms, posing a potential threat to the marine ecosystem.^[4] Therefore, non-toxic and efficient anti-fouling coatings without heavy-metal biocides must be developed.

Extracting or synthesizing natural biocides such as sesquiterpenes,^[5] steroids,^[6] alkaloids,^[7] indoles,^[8] capsaicin,^[9] and their derivatives,^[10] from various organisms, is an effective strategy for replacing heavy-metal

biocides. However, these natural biocides and their derivatives are primarily mixed with resins via physical blending, which leads to their rapid leaching during application and reduces their anti-fouling performance.^[11] To address this issue, the molecular structures of natural biocides have been modified by grafting them onto resin side chains while retaining their anti-fouling-active functional groups.^[12] However, natural biocides and their derivatives exhibit limited repellency against certain bacteria and diatoms.^[13] Anti-fouling properties based solely on natural biocides and their derivatives are insufficient to cope with the complex marine environment; moreover, improving the broad-spectrum anti-fouling properties of coatings is imperative.^[14] Therefore, such natural biocides and their derivatives are combined with materials such as low-surface-energy (SE) coatings,^[15] amphiphilic polymers,^[16] and hydrogels^[17] to obtain a composite coating that exhibits synergistic dual anti-fouling effects. However, during the design of such composite coatings, focus is often placed on improving their anti-fouling performance and the retention of self-polishing performance is neglected.^[18] As fouling organisms adhere and accumulate on the surface of composite coatings, these coatings gradually lose their anti-fouling ability owing to their inability to hydrolyze and self-polish, preventing the formation of a self-renewing surface.^[19] Consequently, the development of a coating that combines satisfactory erosion performance and anti-fouling properties remains challenging.

In this study, a novel self-polishing anti-fouling coating that integrated two anti-fouling mechanisms was synthesized

1. Introduction

Marine fouling organisms can adhere to the underwater surfaces of ships and marine facilities, thereby reducing the navigation speed, increasing fuel consumption, and accelerating the corrosion of such facilities.^[1] Self-polishing copolymers containing heavy-metal biocides are considered highly effective anti-fouling

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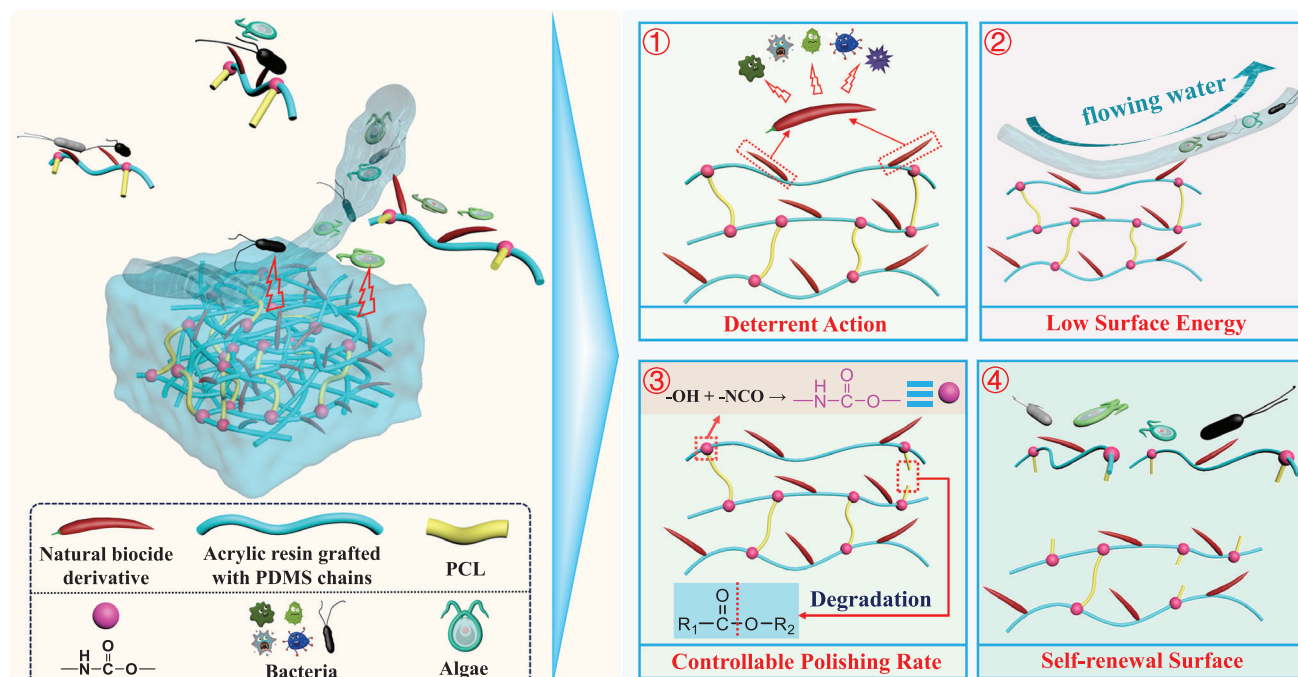
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Scheme 1. Distinctive properties of the self-polishing coating with dual anti-fouling strategies.

(Scheme 1). The challenge of limited broad-spectrum anti-fouling performance in coatings using a single anti-fouling strategy is effectively addressed by combining two anti-fouling strategies: natural anti-fouling agent derivatives and low surface energy. In addition, introducing a degradable polycaprolactone (PCL) chain addresses the issue of compromising self-polishing performance in high-performance anti-fouling coatings, thereby ensuring the stability of their anti-fouling properties. First, a natural biocide derivative, N-(4-hydroxy-3-methoxybenzyl) (HMBA), that could repel fouling organisms was synthesized and immobilized on the side chain of an acrylic resin via radical polymerization (Scheme 10). During polymerization, the polydimethylsiloxane (PDMS) chain was grafted onto the side chains of the acrylic resin, and its low-SE properties were used to further inhibit the adhesion of fouling organisms on the resin surface (Scheme 12). A degradable PCL chain was used to bridge acrylic resin and construct a degradable cross-linked network in seawater, thereby ensuring the self-polishing characteristics of the coating and maintaining its durable anti-fouling ability (Scheme 14). Moreover, this cross-linked network ensured the mechanical stability of the coating (Scheme 13). Compared with the commercial B3000 anti-fouling coating, the as-synthesized anti-fouling coating reduced the adhesion of *Staphylococcus aureus*, *Vibrio alginolyticus* and *Nitzschia sp.* by $\approx 89.5\%$, $>99.9\%$ and $\approx 87.3\%$, respectively.

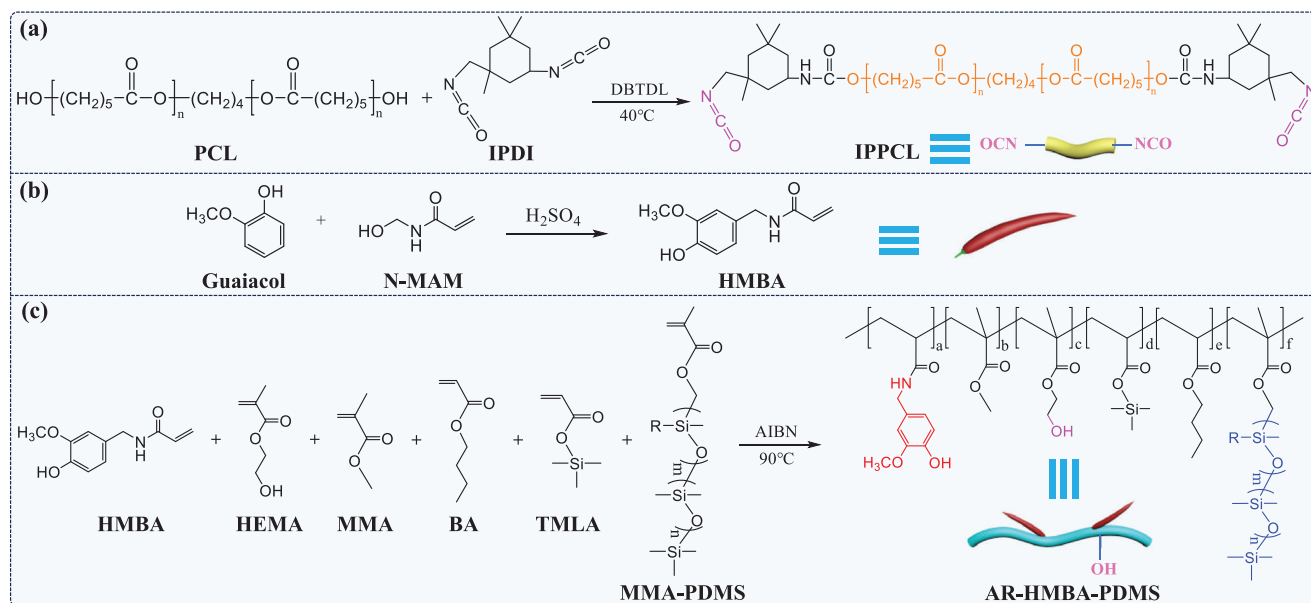
2. Results and Discussion

Scheme 2 outlines the synthesis of isocyanate-terminated polycaprolactone (IPPCL), HMBA, acrylic resin with HMBA and PDMS chains grafted on the side chain of the acrylic resin (AR-

HMBA-PDMS) and the anti-fouling coating (PCLAR-HMBA-PDMS). The detailed synthesis steps and specific ratios are provided in the Supporting Information. First, IPPCL was successfully synthesized by reacting the isocyanate group of isophorone diisocyanate (IPDI) with the alcohol hydroxyl group of PCL (Scheme 2a). Then, HMBA was synthesized via Friedel-Crafts reaction (Scheme 2b). Then, it was uniformly mixed with methyl methacrylate (MMA), butyl acrylate (BA), acryloyloxytrimethylsilane (TMLA), hydroxyethyl methacrylate (HEMA), methyl methacrylate terminated silicone oil (MMA-PDMS) and AR-HMBA-PDMS via free-radical polymerization (Scheme 2c). Among them, MMA and BA were used to regulate the mechanical properties of the coating. TMLA-containing silicone ester groups accelerated the breakage of PCL chains in the coating,^[20] and HEMA-containing hydroxyl groups reacted with IPPCL to form urethane groups.^[21] Finally, AR-HMBA-PDMS was mixed evenly with IPPCL to prepare an anti-fouling coating that was labeled as PCLAR-HMBA-PDMS (Scheme 1). The same synthesis steps were used to prepare PCLAR (without HMBA and PDMS), PCLAR-HMBA (without PDMS), and PCLAR-PDMS (without HMBA). To demonstrate the controllability of IPPCL on the erosion rate of coatings, AR (without adding IPPCL) and PCL₂AR (the amount of IPPCL added is twice that of PCLAR) were also prepared.

2.1. Characterization of IPPCL, HMBA, AR-HMBA-PDMS, and PCLAR-HMBA-PDMS

Table 1 shows the molecular weights and distributions of PCL and IPPCL, which indicates that the dispersion coefficient of IPPCL is relatively narrow and its molecular weight is close to the



Scheme 2. Synthesis process of the (a) IPPCL, b) HMBA, and (c) AR-HMBA-PDMS.

Table 1. Molecular weights and distributions of PCL, IPPCL, AR, AR-HMBA, AR-PDMS, AR-HMBA-PDMS.

Sample	M_w [g·mol ⁻¹]	PDI
PCL	490	1.05
IPPCL	1139	1.15
AR	4948	2.11
AR-HMBA	5233	1.84
AR-PDMS	5432	1.87
AR-HMBA-PDMS	5579	1.66

theoretical value. **Figure 1a** shows the ¹H NMR spectrum of PCL before and after its reaction with IPDI. A broad peak appeared at 4.5 ppm in the spectrum, corresponding to the active hydrogen in the hydroxyl group of PCL. However, this active hydrogen signal was absent in IPPCL and was replaced by peaks at 7.02 and 4.72 ppm corresponding to the active hydrogen related in the -NH-

group. These results demonstrated the complete conversion of the hydroxyl groups in PCL to urethane groups by reacting with IPDI. Moreover, IPPCL showed an N-H stretching vibration peak at 3330 cm⁻¹,^[22] accompanied by corresponding C-N stretching and N-H bending vibration peaks at 1515 cm⁻¹,^[23] confirming the formation of urethane bonds (Figure S1, Supporting Information). Notably, the presence of a characteristic peak at 2267 cm⁻¹ indicated the presence of excess isocyanate groups in the product,^[24] which can participate in the subsequent cross-linking reaction.

The synthesis process of HMBA closely resembled that outlined in a previous with some modifications.^[25] The ¹H NMR spectrum (Figure 1b) shows resonance peaks corresponding to the protons in HMBA, aligning in position and quantity with a previous study, thereby confirming the successful synthesis of HMBA. ¹H NMR (DMSO, ppm): 8.85 (s, 1H, O-H), 8.45 (s, 1H, N-H), 6.83 (s, 1H, -CH-aromatic), 6.71 (s, 1H, -CH-aromatic), 6.66 (d, 1H, -CH-aromatic), 6.26 (d, 1H, -CH=), 5.85 (m, 2H, CH₂=), 4.23 (d, 2H, -CH₂-), 3.73 (s, 3H, O-CH₃).

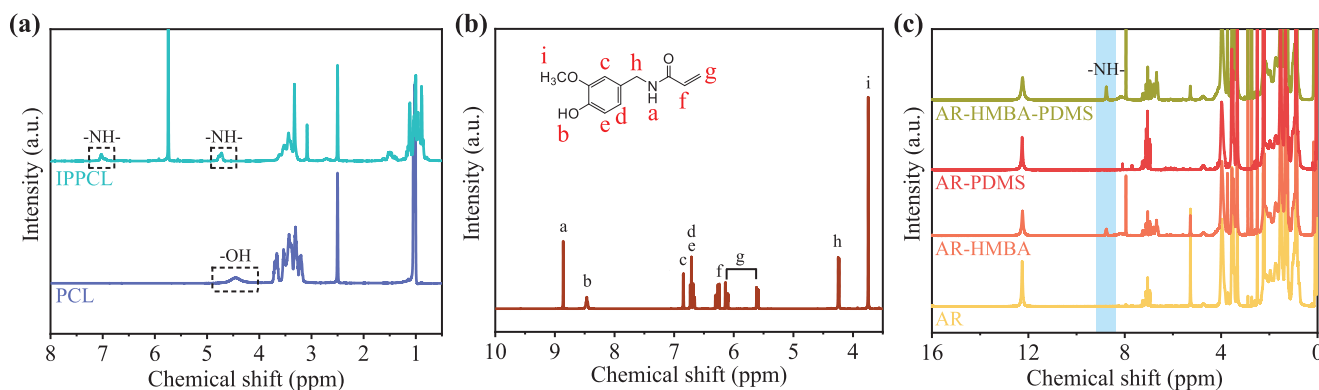


Figure 1. ¹H NMR spectra of (a) PCL and IPPCL, b) HMBA, and (c) AR, AR-HMBA, AR-PDMS and AR-HMBA-PDMS.

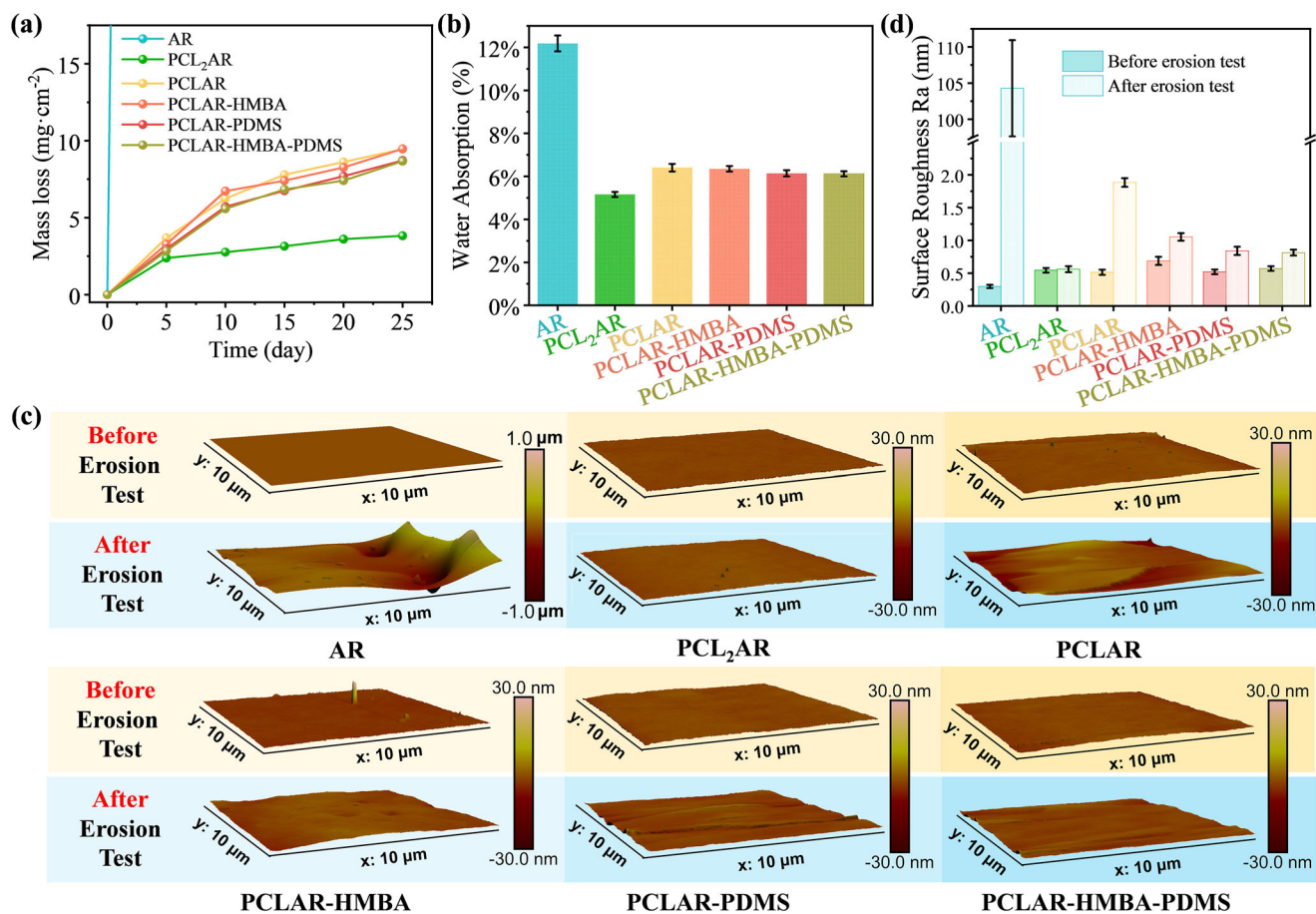


Figure 2. a) Erosion rate curves and (b) water absorption of the coatings; c) surface topographies and (d) surface roughness values of coatings before and after erosion test.

Building upon this foundation, an acrylic resin with HMBA and PDMS grafted onto its side chains was synthesized via free-radical polymerization and designated as AR-HMBA-PDMS. Concurrently, control samples including AR, AR-HMBA, and AR-PDMS were synthesized. As shown in Table 1, the introduction of HMBA and PDMS chains increases the molecular weight and reduces the polydispersity index (PDI) of acrylic resin. Figure 1c shows the ¹H NMR spectrum of acrylic resins containing various components. Notably, AR-HMBA and AR-HMBA-PDMS exhibited discernible differences from AR, evidenced by a proton peak at 8.76 ppm corresponding to the -NH- group; this observation confirmed the ability of HMBA to co-polymerize with other acrylic monomers. Figure S2 (Supporting Information) shows the FT-IR spectrum of acrylic resins, wherein all samples display a prominent peak at $\approx 2959\text{ cm}^{-1}$,^[26] corresponding to the stretching vibration of hydroxyl groups. It is worth noting that the absorption band at 1671 cm^{-1} can be attributed to the stretching mode of the carbonyl group (C=O) in the acid amide of HMBA, which also confirmed the ability of HMBA to co-polymerize with other acrylic monomers. This indicates that acrylic resins can participate in subsequent cross-linking reactions.

Figure S3 (Supporting Information) shows the FT-IR spectrum of coatings. In the spectrum, the characteristic absorption

peak associated with the isocyanate group at 2267 cm^{-1} disappeared and was replaced by an N-H stretching vibration peak at 1515 cm^{-1} . This shift indicates the complete reaction of the isocyanate group from IPPCL with the hydroxyl group in the acrylic resin, thereby forming a urethane bond.

2.2. Erosion Performance of Coatings

PCL chains were used to bridge acrylic acid resin chains and form an anti-fouling coating. The influence of IPPCLs on the erosion of coating was quantitatively assessed using a self-built test device (Figure S4, Supporting Information). As shown in Figure 2a, all coatings exhibited high mass loss in initial testing stages, which gradually stabilized over time. Notably, AR exhibited a remarkable mass loss of $61.946\text{ mg}\cdot\text{cm}^{-2}$ within 1 day. In contrast, PCLAR and PCL₂AR experienced mass losses of 9.456 and $3.824\text{ mg}\cdot\text{cm}^{-2}$ after 25 days, respectively, because the chemical bonding between IPPCL and AR constrained the mobility of AR chains. During erosion, AR necessitated surface chain stripping predominantly via overcoming intermolecular force. In contrast, the intermolecular forces between the surface chains and the hydrolysis of ester groups within IPPCL chains must be overcome to ensure surface chain stripping in PCLAR and PCL₂AR.

By increasing the addition amount of IPPCL, more IPPCL chains will be attached to the same AR chain and more ester groups will be hydrolyzed. Furthermore, the mass losses of PCLAR-HMBA, PCLAR-PDMS, and PCLAR-HMBA-PDMS after 25 days were 9.472, 8.72, and 8.672 mg·cm⁻², respectively. The erosion rate of coatings was only marginally impacted after the addition of HMBA and slightly reduced after the addition of PDMS.

The relation between the erosion rate and water absorption was determined by analyzing the water absorption rates of different coatings. As shown in Figure 2b, AR, PCL₂AR, PCLAR, PCLAR-HMBA, PCLAR-PDMS, and PCLAR-HMBA-PDMS exhibited water absorption rates of 12.19%, 5.16%, 6.41%, 6.35%, 6.14%, and 6.12%, respectively. Moreover, a positive correlation existed between water absorption and erosion rate of coatings. Evidently, the formation of a cross-linking network via the chemical bonding of IPPCL and AR substantially impeded the penetration of water molecule and reduced the water absorption of coatings as the cross-linking degree increased.^[27] Simultaneously, the addition of PDMS chains enhanced the hydrophobicity of coatings (Section 2.4) and reduced the water absorption of coatings.^[28]

The smooth surface of the anti-fouling coating is crucial for reducing frictional resistance during voyage, which can decrease fuel consumption and ensure navigation stability.^[29] Thus, we captured macroscopic images of the coating before and after the erosion test (Figure S5, Supporting Information). All coating surfaces were smooth and flat before erosion test; however, visible wrinkles appeared on the AR surface after the erosion test, and the macroscopic morphology of the other coatings remained largely unchanged. Atomic force microscopy was performed to obtain surface morphology and roughness data for elucidating the microscopic changes on the coating surface (Figure 2c,d). The surface roughness of AR, PCL₂AR, PCLAR, PCLAR-HMBA, PCLAR-PDMS, and PCLAR-HMBA-PDMS were 0.30, 0.54, 0.51, 0.69, 0.52, and 0.57 nm, respectively, before the erosion test, indicating their uniform and smooth surfaces. However, the surface roughness of coatings slightly increased after the erosion test to 104.28, 0.56, 1.88, 1.05, 0.84, and 0.81 nm, respectively, because they adsorbed water and swelled subsequently, followed by dehydration and shrinkage.^[30] Notably, the surface roughness of AR increased considerably, indicative of its relatively weak mechanical stability, prone to forming wrinkles under seawater scouring. In contrast, the surface roughness of remaining coatings increased slightly after the erosion test with negligible morphological changes. This indicated that these coatings maintained a relatively smooth and flat surface, which is beneficial for marine applications.

2.3. Mechanical Properties of the Coatings

Anti-fouling coatings are the outermost layers of marine protective coatings that are subjected to substantial loads during maritime navigation; therefore, they should have robust mechanical stability.^[31] To elucidate the underlying mechanisms that caused changes in morphology and roughness of coatings during the erosion test, the influence of cross-linking degree and coating composition on their mechanical properties was investigated via tensile tests. The tensile strengths of AR, PCL₂AR, and PCLAR were 0.15, 1.60, and 0.84 MPa, respectively (Figure 3a).

The cross-linking degree increased with increasing IPPCL content, thereby increasing the number of effective chains that can withstand deformation loads and enhance the tensile strength.^[32] The elongation at break values for AR, PCL₂AR, and PCLAR were 0.03%, 2.99%, and 0.34%, respectively. The enhanced cross-linking density unexpectedly increased the elongation at break (Figure 3b). This can be attributed to the low steric hindrance of the methylene group of the IPPCL chain, which provided flexibility to the coating.^[33] The incorporation of HMBA slightly influenced the tensile strength and elongation at break. The tensile strength and elongation at break of PCLAR-HMBA were 0.97 MPa and 0.34%, respectively, and those of PCLAR-PDMS and PCLAR-HMBA-PDMS were 1.72 MPa and 1.79 MPa, with corresponding elongation at break of 2.76% and 2.79%, respectively. The addition of PDMS chains enhanced the tensile strength and elongation at break of the coating. The low rotational barrier of the Si—O—Si bond endowed the PDMS chains with remarkable flexibility.^[34] As a result, the flexible polymer chains facilitated stress dispersion and transmission and mitigated the likelihood of stress concentration and local damage, thereby enhancing the tensile strength and elongation at break of the coating.^[35]

Robust adhesion between the coating and substrate is a vital prerequisite for the secure and dependable application of anti-fouling coatings. Thus, the adhesion strength of coatings on epoxy fiberboard surfaces was assessed utilizing an adhesion tester. As shown in Figure 3c, the adhesion strengths of AR, PCL₂AR, and PCLAR were 0.55, 1.14, and 1.01 MPa, respectively, and increased with increasing cross-linking degree. This increment can be attributed to the increase in the number of urethane groups due to the reaction of IPPCL and AR. These urethane groups are rich in hydrogen bonds, which strengthened the adhesion between the coating and substrate.^[36] Thus, PCLAR-HMBA, PCLAR-PDMS, and PCLAR-HMBA-PDMS exhibited adhesion strengths of 1.58, 0.75, and 1.2 MPa, respectively. Upon adding HMBA, the adhesion of coatings enhanced due to the presence of amide group (—CONH₂) that facilitated the interaction with polar groups on the substrate surface and improved adhesion.^[37] However, the introduction of PDMS chains marginally diminished the coating adhesion owing to their hydrophobicity.

Thus, the cross-linking degree of coatings is crucial for enhancing the erosion rate and mechanical stability of coatings. The contribution of cross-linking network structure to the coating stability was obtained using the differential curve data via thermogravimetric analysis.^[38] The enlarged inset in Figure 3d depicts the thermal decomposition behavior of coatings at 300–500 °C. Notably, the maximum thermal decomposition temperatures of AR, PCL₂AR, and PCLAR were 385.16, 406.5, and 403.83 °C, respectively. The maximum thermal decomposition temperature increased with increasing cross-linking degree of coating, indicating that the cross-linking network structure formed by IPPCL and AR effectively retarded the bond-breaking process and enhanced the stability of coatings.^[39] The microstructural changes in coatings were observed from their high-resolution N 1s spectra (Figure 3e). Within the investigated energy spectrum range, AR exhibited no significant N 1s peak. In contrast, both PCLAR and PCL₂AR exhibited a singular peak at ≈400.1 eV, corresponding to the C—N bond in the urethane group.^[40] Notably, the N 1s peak intensity of PCL₂AR was higher than that of PCLAR, possibly

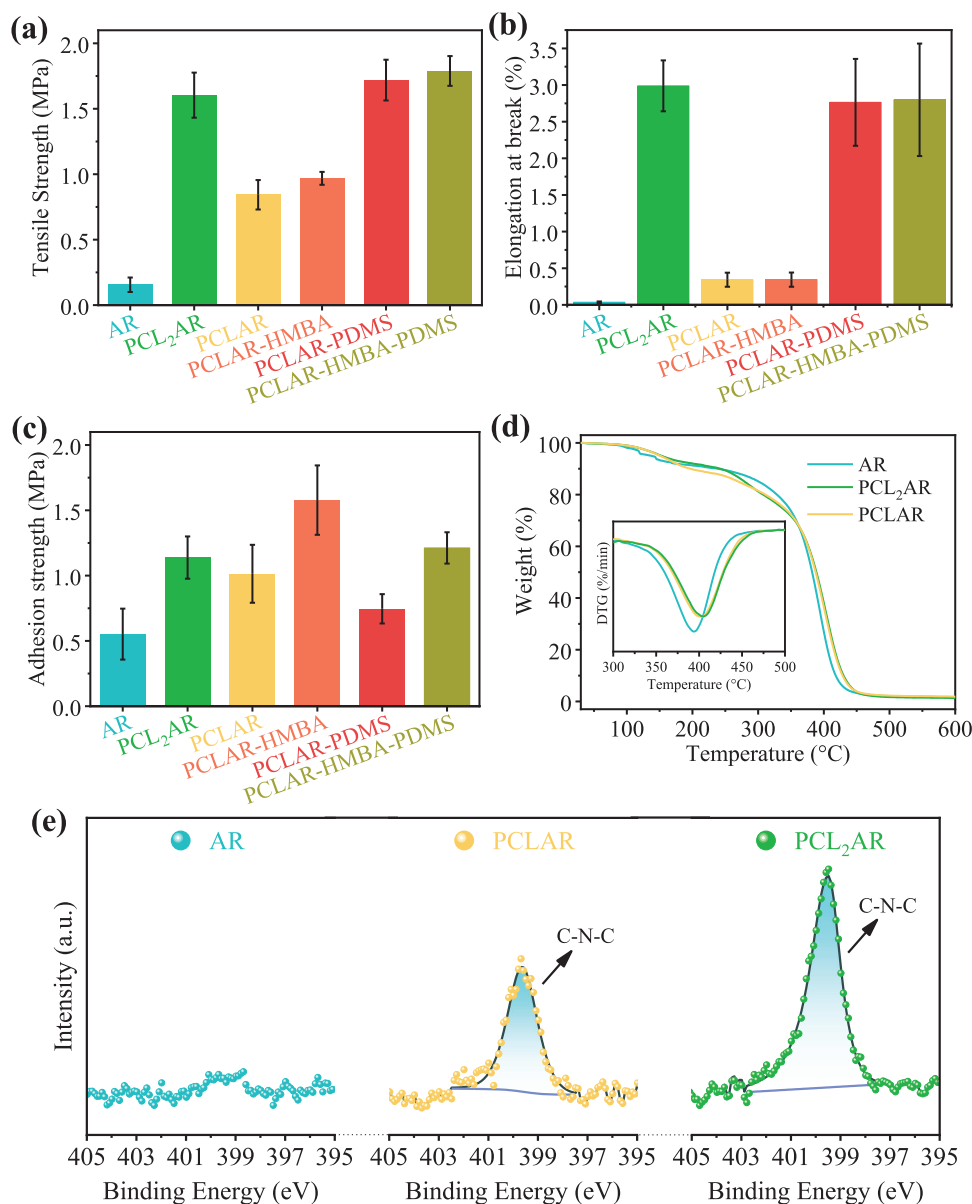


Figure 3. a) Tensile strength, b) elongation at break, c) adhesion strength, d) differential thermogravimetric curves of the coatings, and e) high-resolution N 1s spectra of the coatings.

due to the increased number of urethane bonds resulting from increased IPPCL content. The C–N group in HMBA molecules enhanced the N 1s peak intensity at ≈ 400.1 eV for PCLAR-HMBA and PCLAR-HMBA-PDMS (Figure S6, Supporting Information).

For their practical application, anti-fouling coatings must have appropriate erosion rates and robust mechanical stability. AR exhibits an excessively rapid erosion rate coupled with poor mechanical stability, resulting in excessive wear and increasing the production costs and resource consumption. PCL₂AR also exhibited considerably lower erosion rates, rendering it susceptible to the accumulation of fouling organisms and gradual loss of its anti-fouling ability. Therefore, we will focus on assessing the anti-fouling performance of PCLAR, PCLAR-HMBA, PCLAR-PDMS, and PCLAR-HMBA-PDMS.

2.4. Anti-Fouling Performance

Marine fouling organisms attach directly on the surface of coatings, and surface characteristics highly influence this behavior of organisms. To accurately understand the influence of surface characteristics on anti-fouling performance, the hydrophilic, hydrophobic, oleophilic, and oleophobic properties of coatings were comprehensively assessed by determining the static contact angles of water, kerosene, and n-hexadecane on the coating surface (Figure 4a–c). Based on the measured values, the OWRK method was used to accurately calculate the SE of coatings (Figure 4d). Figure S7 (Supporting Information) shows the outlines of various liquid media on the coating surface. As shown in Figure 4a, PCLAR shows a hydrophilic state with a static

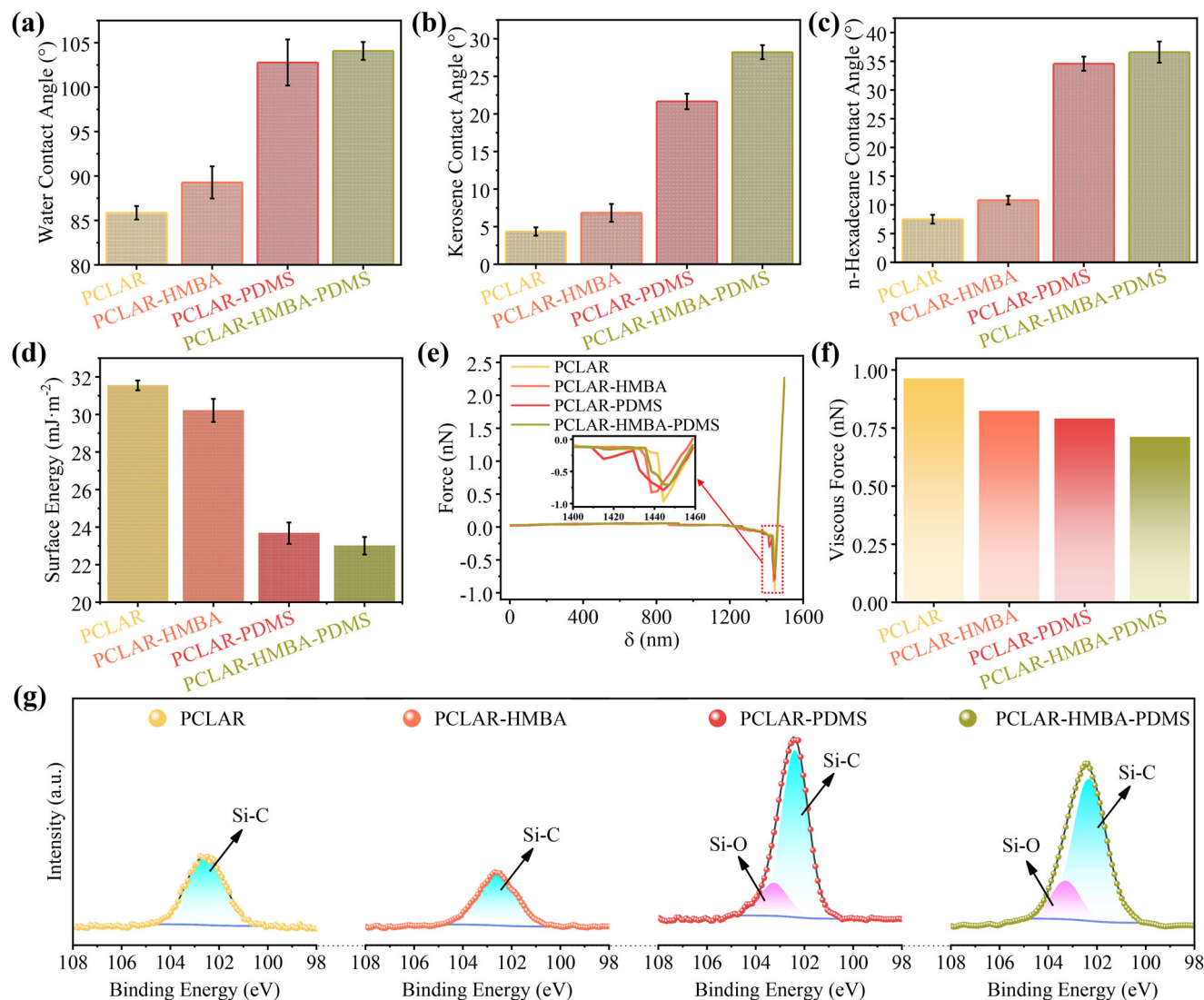


Figure 4. Static contact angles of (a) water, (b) kerosene and (c) n-hexadecane on the coating surface; d) SEs, (e) force-displacement curves and (f) viscous forces of the coating surface and (g) high-resolution Si 2p XPS spectra of the coatings.

water contact angle of $\approx 85.9^\circ$. The static water contact angle slightly increased to 89.3° upon adding HMBA, indicating a modest enhancement in its hydrophobicity. The addition of PDMS chains considerably enhanced the hydrophobicity of PCLAR-PDMS and PCLAR-HMBA-PDMS, resulting in static water contact angles of 102.8° and 104.1° , respectively. The static contact angles of kerosene (Figure 4b) and n-hexadecane (Figure 4c) were similar to that of water. In particular, PCLAR exhibited static kerosene and n-hexadecane contact angles of 4.4° and 7.5° , which slightly increased to 6.8° and 10.8° , respectively, upon adding HMBA. PDMS chains considerably enhanced the static contact angles of kerosene and n-hexadecane on PCLAR-PDMS and PCLAR-HMBA-PDMS coatings to 21.7° and 34.6° as well as 28.2° and 36.6° , respectively. The trend of changes in SE and contact angle was completely contrasting. PCLAR exhibited the highest SE of $\approx 31.6 \text{ mJ}\cdot\text{m}^{-2}$, which marginally reduced to $30.2 \text{ mJ}\cdot\text{m}^{-2}$ upon adding HMBA (PCLAR-HMBA). Notably, the ad-

dition of PDMS chains considerably reduced the SEs of PCLAR-PDMS and PCLAR-HMBA-PDMS to 23.7 and $23.0 \text{ mJ}\cdot\text{m}^{-2}$, respectively.

To further elucidate the interaction between the coating and fouling organisms, the force-distance curve of the coatings was obtained via atomic force microscopy.^[41] As shown in Figure 4e, the indentation and retraction curves of all coatings do not overlap, indicating the presence of a significant viscous force between the coating and probe. Typically, the magnitude of this viscous force reflects the frictional resistance encountered during the contact and relative motion between the coating and probe, as well as the difficulty in detaching objects from the coating surface. Figure 4f shows the specific values of the viscous force, with PCLAR exhibiting the highest value of 0.9638 nN . Upon introducing HMBA and PDMS, the viscous force decreased. Specifically, the viscous forces of PCLAR-HMBA, PCLAR-PDMS, and PCLAR-HMBA-PDMS were 0.8241 , 0.7911 , and 0.712 nN ,

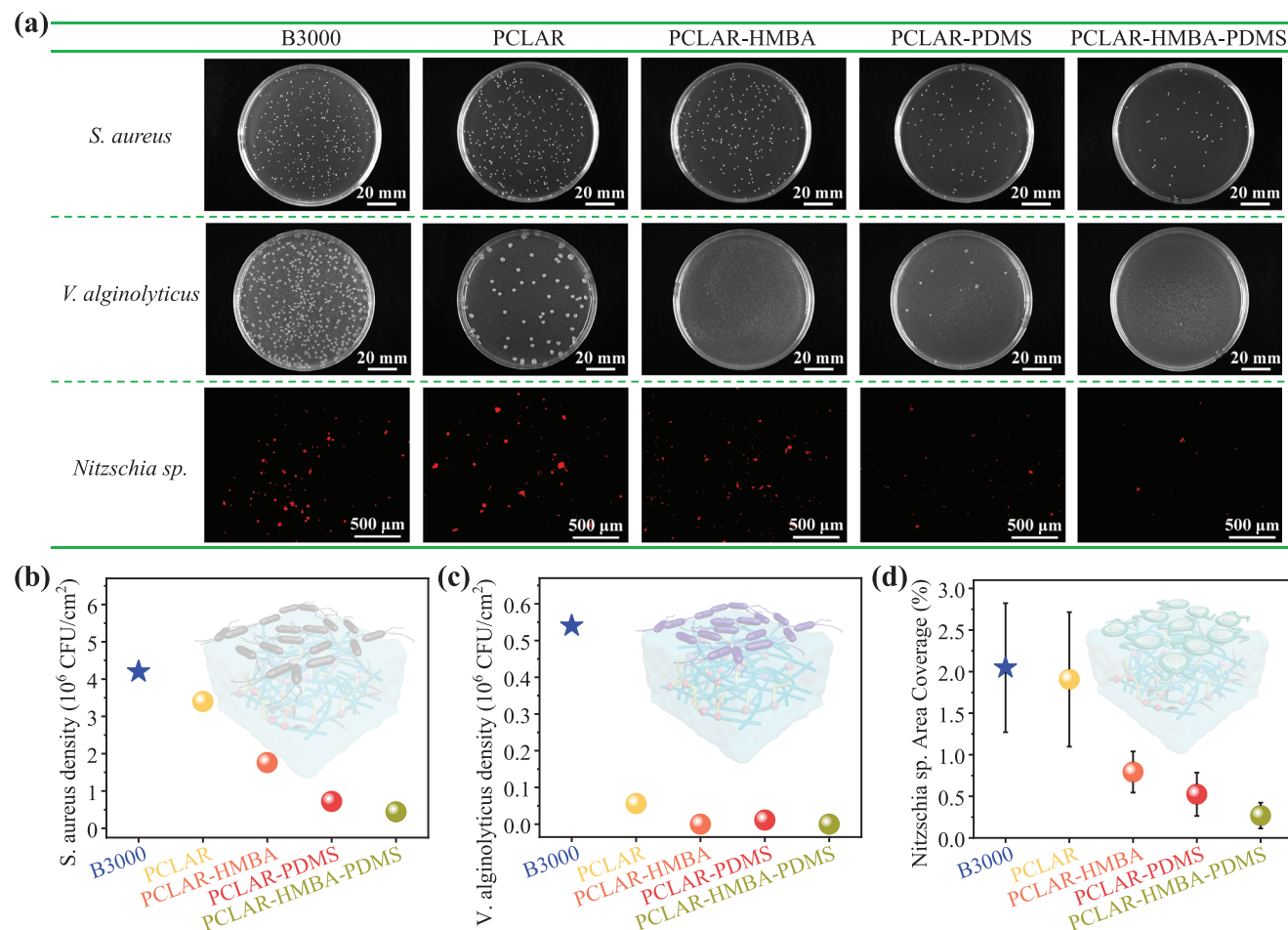


Figure 5. a) Distributions of *S. aureus* and *V. alginolyticus* on the LB medium; *Nitzschia sp.* cell distribution on the coating surfaces. Bacterial densities of (b) *S. aureus* and (c) *V. alginolyticus* on the LB medium after incubation with different coatings for 24 h. d) Area coverage of *Nitzschia sp.* on the coating surfaces.

respectively. The trend observed for the viscous force on the coating surface aligned with that of SE.

To elucidate the intrinsic factors that influenced the changes in SE, the full XPS spectrum (Figure S8, Supporting Information) and high-resolution Si 2p spectrum of the coating was obtained (Figure 4g). The quantitative analysis results for the atomic fractions in the probed surface regions are summarized in Table S2 (Supporting Information). The high-resolution Si 2p XPS spectra of PCLAR and PCLAR-HMBA were fitted to a single peak at 102.3 eV, attributed to the Si-C bond. However, those of PCLAR-PDMS and PCLAR-HMBA-PDMS were deconvoluted into two peaks at 102.3 eV (Si-C) and 103.3 eV (Si-O), respectively.^[42] Moreover, compared with that of PCLAR and PCLAR-HMBA, the Si 2p peak intensity of PCLAR-PDMS and PCLAR-HMBA-PDMS was notably elevated. This indicated that the enrichment of PDMS chains on the coating surface considerably reduced the SE.

The efficacy of marine protective coatings in preventing the adhesion of biofouling organisms is crucial for ensuring the uninterrupted operation of vessels and equipment under harsh marine environments. Previous studies have highlighted the pivotal role of bacterial adhesion on the coating surfaces during the initial stages of marine fouling.^[43] Consequently, *S. aureus*,

a representative Gram-positive strain, and *V. alginolyticus*, a representative Gram-negative strain, were selected to investigate the anti-bacterial adhesion performance of the fabricated coatings. The anti-fouling efficacy of coatings was assessed by testing the anti-bacterial activity using the plate count method and commercially anti-fouling coating B3000 as the blank control. Figure 5a shows the distribution of *S. aureus* on LB medium, and Figure 5b shows the corresponding colony density. The surfaces of B3000 and PCLAR exhibited a large number of *S. aureus* colonies, with colony densities of 4.2×10^6 and 3.4×10^6 CFU·cm⁻², respectively, indicating limited anti-bacterial activity. However, upon introducing HMBA and PDMS chains, the adhesion of *S. aureus* was effectively inhibited and their colony densities on the PCLAR-HMBA and PCLAR-PDMS surfaces decreased to 1.76×10^6 and 0.72×10^6 CFU·cm⁻², respectively. The anti-adhesion efficiencies of PCLAR-HMBA and PCLAR-PDMS against *S. aureus* reached 58.1% and 82.9%, respectively. Notably, the simultaneous introduction of HMBA and PDMS chains further inhibited the adhesion of *S. aureus*, thereby decreasing the colony densities on the PCLAR-HMBA-PDMS surface to 0.44×10^6 CFU·cm⁻² and exhibiting an anti-adhesion efficiency of 89.5%. As shown in Figure 5a, the adhesion of *V. alginolyticus*

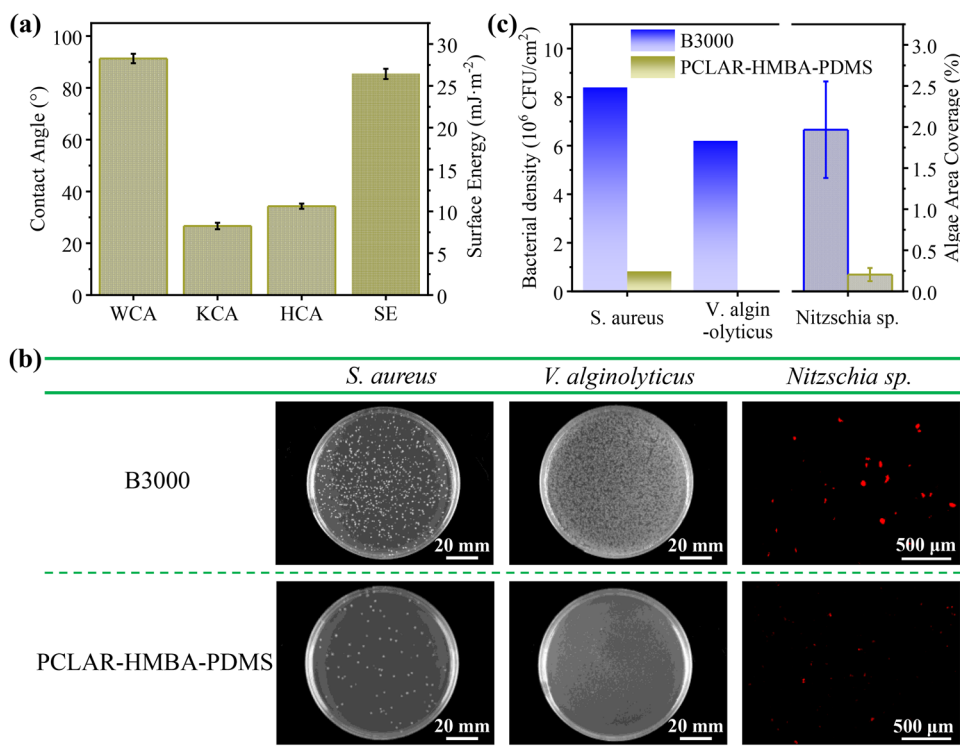


Figure 6. a) Static contact angles of water, kerosene, and n-hexadecane on the PCLAR-HMBA-PDMS surface after the erosion test. b) Distributions of *S. aureus* and *V. alginolyticus* on the LB medium and distributions of *Nitzschia sp.* algal cells on the B3000 and PCLAR-HMBA-PDMS surfaces after the erosion test. c) Bacterial densities of *S. aureus* and *V. alginolyticus* on the LB medium after incubation for 24 h and area coverages of *Nitzschia sp.* on the B3000 and PCLAR-HMBA-PDMS surfaces after the erosion test.

to the coating surface was relatively low; the colony density of 5.2×10^5 CFU·cm⁻² was observed on the B3000 surface, indicating limited anti-bacterial activity. Meanwhile, the colony densities on the PCLAR and PCLAR-PDMS surfaces were 5.6×10^4 and 1.1×10^4 CFU·cm⁻², respectively. The introduction of PDMS chains reduced the adhesion of *V. alginolyticus* to a certain extent and yielded an anti-adhesion efficiency of $\approx 98\%$. Surprisingly, the colony densities on the PCLAR-HMBA and PCLAR-HMBA-PDMS surfaces were $< 1 \times 10^3$ CFU·cm⁻², indicating that the introduction of HMBA considerably inhibited the adhesion of *V. alginolyticus* on the coating surface and yielding an anti-adhesion efficiency of $> 99.9\%$.

Algae are common marine fouling organisms that facilitate adhesion for larger fouling organisms and serve as a nutrient source for their survival.^[44] Therefore, evaluating the anti-algae performance of coatings against diatoms is crucial. Herein, the area coverage of *Nitzschia sp.* on the surface of fabricated coatings was quantitatively analyzed based on their red fluorescence characteristics generated by chlorophyll in their cells. Figure 5a shows the distribution of *Nitzschia sp.* on the coating surface, and Figure 5c shows the specific area coverages. A substantial number of *Nitzschia sp.* adhered to the B3000 and PCLAR surfaces and clustering was observed in some areas, resulting in area coverage rates of 2.04% and 1.96%, respectively; these values indicated the poor anti-algae adhesion properties of the fabricated coatings. In contrast, the area coverage rates of *Nitzschia sp.* on the PCLAR-HMBA and PCLAR-PDMS surfaces were 0.79% and 0.52%, respectively, demonstrating that the in-

roduction of HMBA and PDMS chains effectively inhibited its adhesion. Moreover, the anti-algae adhesion efficiencies of these coatings were $\approx 61.3\%$ and 74.5% , respectively. The combination of HMBA and PDMS chains further reduced the adhesion of *Nitzschia sp.* on the PCLAR-HMBA-PDMS surface, and its area coverage rate reduced to 0.26% and the anti-algae adhesion efficiency increased up to 87.3%.

2.5. Stability of Anti-Fouling Performance

The fabricated coatings were accompanied by the degradation and breakage of PCL chains during their application. Therefore, their surface characteristics must be analyzed in depth after the erosion test to evaluate their anti-fouling performance. PCLAR-HMBA-PDMS exhibited the best anti-fouling performance in previous studies; therefore, the evolution in its performance after the erosion test was particularly analyzed herein. As shown in Figure 6a, the static contact angles of water, kerosene, and n-hexadecane on the coating surfaces decreased to 91.3° , 26.7° , and 34.3° , respectively. Moreover, the SE slightly increased to 26.4 mJ·m⁻². Figure S9 (Supporting Information) shows the full XPS spectrum of PCLAR-HMBA-PDMS after the erosion test, while Table S3 (Supporting Information) presents a summary of the quantitative analysis results for the atomic fractions in the probed surface regions. This XPS spectrum displays the characteristic peak of Na 1s in addition to the characteristic peaks of C 1s, N 1s, O 1s, Si 2s, and Si 2p. Figure S10 (Supporting Information) shows

the high-resolution O 1s spectra of PCLAR-HMBA-PDMS before and after the erosion test. The high-resolution O 1s XPS spectra of the PCLAR-HMBA-PDMS surface before the erosion test are deconvoluted into two peaks at 533.50 eV (C=O) and 532.20 eV (C–O). Meanwhile, those after the erosion test are deconvoluted into three peaks at 535.81 (–OH), 533.88 (C=O), and 532.34 eV (C–O).^[45] The characteristic peak of Na and the peak at 535.81 eV indicate the breakage of ester groups and the formation of hydroxyl and carboxyl groups, which decreased the static contact angles of liquids on the coating surface and increased the SE. Figure S11 (Supporting Information) shows the FT-IR spectrum of PCLAR-HMBA-PDMS after the erosion test. The broad peak observed at 3367 cm^{–1} corresponds to the stretching vibrations of the –OH group,^[26] implying the hydrolysis degradation of the ester groups.

Figure 6b,c shows the anti-fouling performance of PCLAR-HMBA-PDMS and B3000 after the erosion test. B3000 exhibited a high number of *S. aureus* and *V. alginolyticus* colonies after the erosion test, with colony densities of 8.4×10^6 and 6.2×10^6 CFU·cm^{–2}, respectively. Meanwhile, the surface of PCLAR-HMBA-PDMS exhibited remarkable anti-bacterial adhesion properties, with colony densities of 0.82×10^6 CFU·cm^{–2} (*S. aureus*) and $<1 \times 10^3$ CFU·cm^{–2} (*V. alginolyticus*), respectively. The area coverages of *Nitzschia* sp. on the surfaces of B3000 and PCLAR-HMBA-PDMS were 1.97% and 0.20%, respectively. Moreover, the anti-adhesion efficiencies of PCLAR-HMBA-PDMS against *S. aureus* and *V. alginolyticus* reached 90.2% and >99.9%, respectively, and the anti-algae adhesion efficiency was 89.8% after the erosion test. These results show that PCLAR-HMBA-PDMS exhibited excellent anti-fouling performance after the erosion test.

2.6. PCLAR-HMBA-PDMS Performance Analysis

The aforementioned research results show that PCLAR-HMBA-PDMS exhibits excellent comprehensive performance. Mechanical testing and atomic force microscopy imaging measurement results show that PCLAR-HMBA-PDMS can maintain satisfactory surface roughness after erosion. This can be attributed to the cross-linking network formed by the reaction between IP-PCL and AR, which effectively limits chain mobility and ensures coating mechanical stability, enabling the coating to withstand deformation loads. By leveraging the degradability of ester groups in the IPPCL chains in seawater, the coatings achieved a dynamically self-renewing surface under seawater scouring. Anti-fouling test results indicate that the incorporation of PDMS chains and HMBA enhances the anti-fouling performance of the coating. XPS analysis reveals that the accumulation of PDMS chains on the surface substantially reduced the SE of the coating, which is attributed to its excellent liquid repellency. Fouling organisms generally cannot easily adhere to a coating surface with low SE;^[41] thus, the incorporation of PDMS chains effectively inhibits bacterial and algal adhesion. In contrast to the incorporation of PDMS chains, the incorporation of HMBA negligibly affects the hydrophilic–hydrophobic properties of the coating. However, HMBA incorporation into the coating endows the coating with strong anti-bacterial and anti-algal adhesion properties, likely owing to the repellency of HMBA toward fouling

organisms.^[15a] This repellency of HMBA exhibits synergism with the low-SE characteristics provided by the PDMS chain, making PCLAR-HMBA-PDMS the most effective anti-fouling formulation. PCLAR-HMBA-PDMS maintains strong anti-fouling properties even after the erosion test. This is owing to the degradation of PCL chains in seawater, which enables the coating to continuously form new surfaces under erosion, thereby ensuring the stability of anti-fouling performance.

3. Conclusion

A self-polishing anti-fouling coating with dual anti-fouling strategies was developed by integrating HMBA and PDMS chains into the side chains of acrylic resin along with degradable PCL to cross-link the resin. The repellency of HMBA effectively inhibited the adhesion of fouling organisms, while its chemical fixation on the resin side chain ensured long service life and minimal leakage, contributing to marine environmental protection. The enrichment of PDMS chains on the coating surface enhanced liquid repellency, thereby reducing the SE to 23.0 mJ·m^{–2}. Using the synergistic anti-fouling mechanism of low SE and HMBA, the coating exhibited enhanced anti-bacterial and anti-algae adhesion properties. The anti-adhesion efficiencies of PCLAR-HMBA-PDMS against *S. aureus*, *V. alginolyticus*, and *Nitzschia* sp. reached 89.5%, >99.9%, and 87.3%, respectively. The dual anti-fouling mechanism effectively addresses the limitations of broad-spectrum anti-fouling performance typically found in coatings that rely on a single anti-fouling strategy. Furthermore, degradable PCL chains were incorporated into the cross-linked network of the coating. As PCL degrades in seawater, the coating continuously renews its surface under seawater scouring, with the erosion rate controlled by the degree of cross-linking. Therefore, the coating maintains excellent anti-fouling performance after erosion testing. The incorporation of a degradable cross-linked network structure successfully overcomes the challenge of achieving self-polishing characteristics, which are often neglected in many anti-fouling coatings that focus solely on performance. This ensures long-term stability of the anti-fouling properties of the coating. This innovative design not only paves the way for the development of environmentally friendly, high-performance, and durable anti-fouling coatings but also provides valuable insights and strategies for future anti-fouling research.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords

anti-fouling, erosion, low surface energy, mechanical stability, natural biocide derivative

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